# NEUROFUSION AI | PRELIMINARY AI ANALYSIS REPORT

**REPORT ID: REP-20261102-EEE46B**

Automated Multi-Modal AI Results • Stage 1: Pending Clinical Sign-Off

November 02, 2026 • 16:42 UTC

| PATIENT INFORMATION | ATTENDING NEUROLOGIST | EXAMINATION DETAILS |
|---|---|---|
| **ID:** PAT-1083<br>**Name:** Eleanor Vance<br>**Age/Gender:** 67 yrs / Female<br>**Status:** Parkinson's Disease (Hoehn & Yahr Stage II) | **Name:** Dr. Clara Rodriguez, MD<br>**License:** MD-NEURO-77491<br>**Dept:** Neurology & Movement Disorders<br>**Hospital:** Metropolitan Neurological Inst. | **Scan Date:** 2026-11-02<br>**Report Date:** 2026-11-02<br>**Modality:** Mri<br>**Device/Protocol:** NeuroFusionAI Protocol PARKINSON v2.0 |

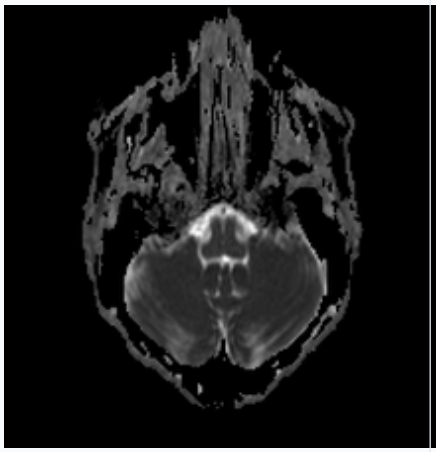
## PRIMARY AI PREDICTION: NORMAL

**Confidence:** 100.0%
**Target:** Parkinson's Disease

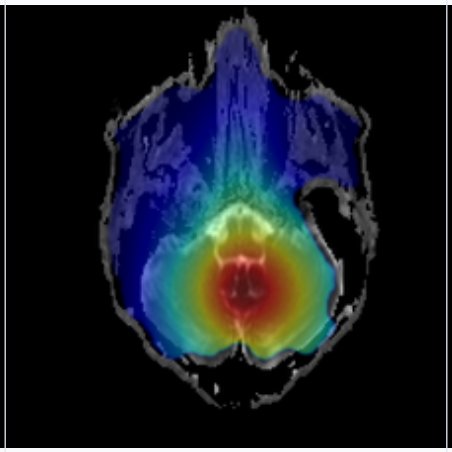
**Class Probability Distribution:** Normal: 100.0% • Parkinson: 0.1%

*AI Confidence Calibrated via Temperature Scaling*

## 1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY



Input MRI Scan (Axial)



Grad-CAM Activation Overlay

**Model:** DenseNet-121 Intracranial Classifier
**Attention Region:** Medial Temporal Lobe & Periventricular Parenchyma
**Intracranial Masking:** Skull-stripped parenchyma segmentation
**Grad-CAM Note:** Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

XAI Interpretation

## 7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

**RISK LEVEL:**
**LOW CLINICAL RISK**

**Urgency:** Routine Outpatient Follow-up

**Evidence-Based Diagnostic Recommendations:**

- Correlate AI automated assessment with comprehensive MDS-UPDRS motor and cognitive exam.
- Recommend DaTscan SPECT dopamine transporter neuroimaging to confirm nigrostriatal deficit.
- Order formal neuropsychological battery (MoCA / MMSE) to track longitudinal cognitive trajectory.
- Initiate outpatient physical therapy and occupational therapy motor rehabilitation.

## PATIENT LONGITUDINAL DIAGNOSTIC HISTORY

| Date | Modality | Prediction | Confidence | Clinical Status |
|---|---|---|---|---|
| 2026-11-02 | Mri | Normal | 100.0% | Verified Match |
| 2026-11-02 | Fusion | Normal | 95.8% | Verified Match |

| | | | | |
|---|---|---|---|---|
| 2026-11-02 | Voice | Parkinson | 89.5% | Verified Match |
| 2026-11-02 | Spiral | Parkinson | 87.4% | Verified Match |

| PRELIMINARY AI SYSTEM OBSERVATIONS: | CLINICAL APPROVAL STATUS |
|---|---|
| Automated vocal biomarker screen shows dysphonia. | *Preliminary AI analysis for decision support. Official physician certification pending.* Status: **AWAITING PHYSICIAN REVIEW & APPROVAL** |

***REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:***

*NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.*